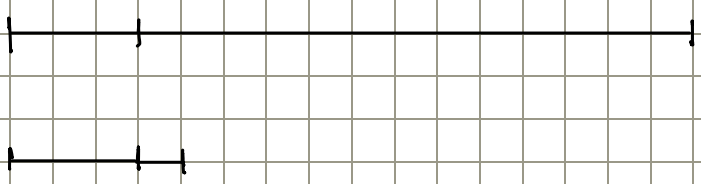
QoS Mechanism

QoS in IP Networks

- Internet: making the best of best effort
- Internet with QoS guarantees:
  - Software defined Networking: differentiated guarantees
  - Network Slicing: firm guarantees (per class traffic) introduced with 5G

- URLCC → industrial processes (hardest to implement with 3G/4G because of reliability and low delay requirements)
- eBB → MM & entertainment
- MTC → IoT



QoS Guarantees

Packet Marking:

Policing: force source adherence to bandwidth allocation
- avoid application send more than what declared

- Marking and policing at network edge

Principle 2

Principle

Tools

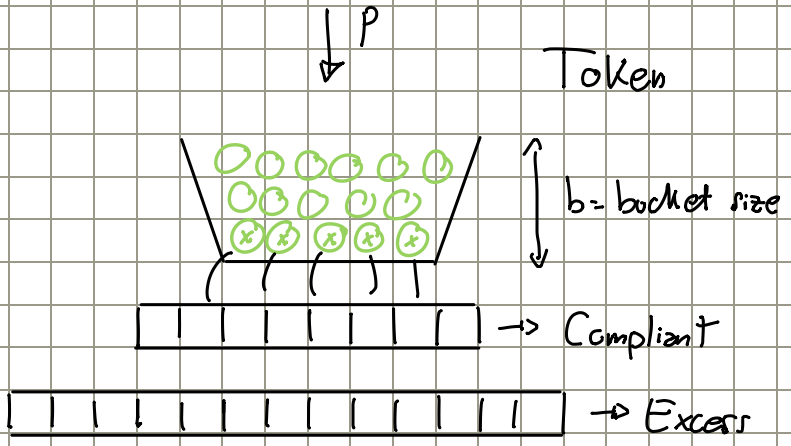
- Classification: grouping traffic based on specific attributes
  How? define classes based on port number and address
- Marking: tagging with QoS labels
- Policing: enforcing bandwidth limits by dropping or remarking traffic exceeding limits
- Shaping: controlling traffic flow to conform to a desired profile buffering non-conformant packets
- RM + Scheduling: determining the allocation of resources and the order of transmission

QoS chain (picture)

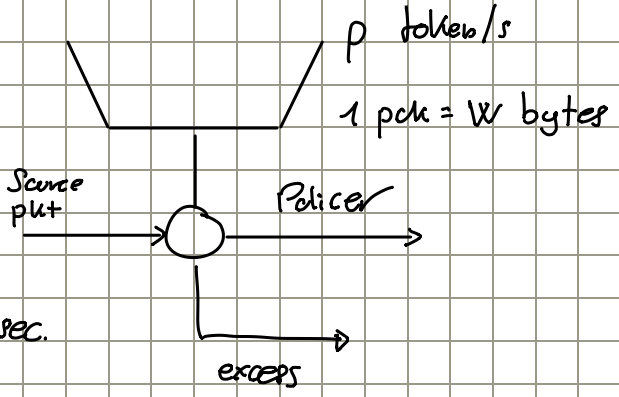
Token Bucket (TB)

Token Bucket Filter

- Bucket holds tokens → permission to transmit data



1) Empty bucket



1 token = 1 byte

1 pck every $\frac{W}{\rho}$ sec.

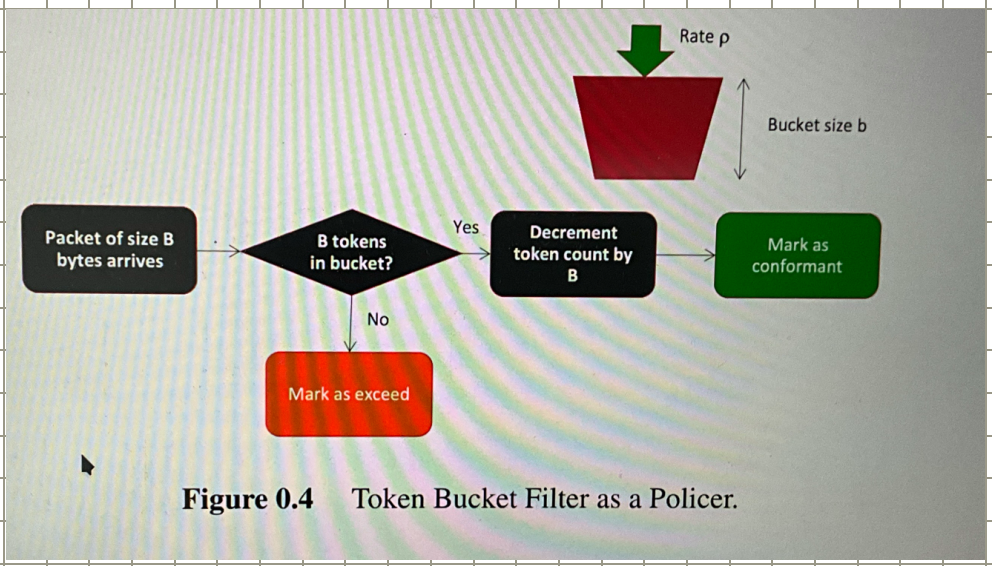


Figure 0.4 Token Bucket Filter as a Policer.

CIR (Committed Information Rate) ρ

ρ = "long term "average" traffic guaranteed by the network

CBS (Committed Bucket Size): b

Max number of tokens that can be collected in the bucket
If determines bucket is full, any additional token will be discarded

Example

CBR: A = 20 ms, B = 1 pck/batch W = 80 byte/packet (Source Model)

TBF params:

1 token = 1 byte, CIR: ρ = 3000 token/s CBS: b = 250 tokens

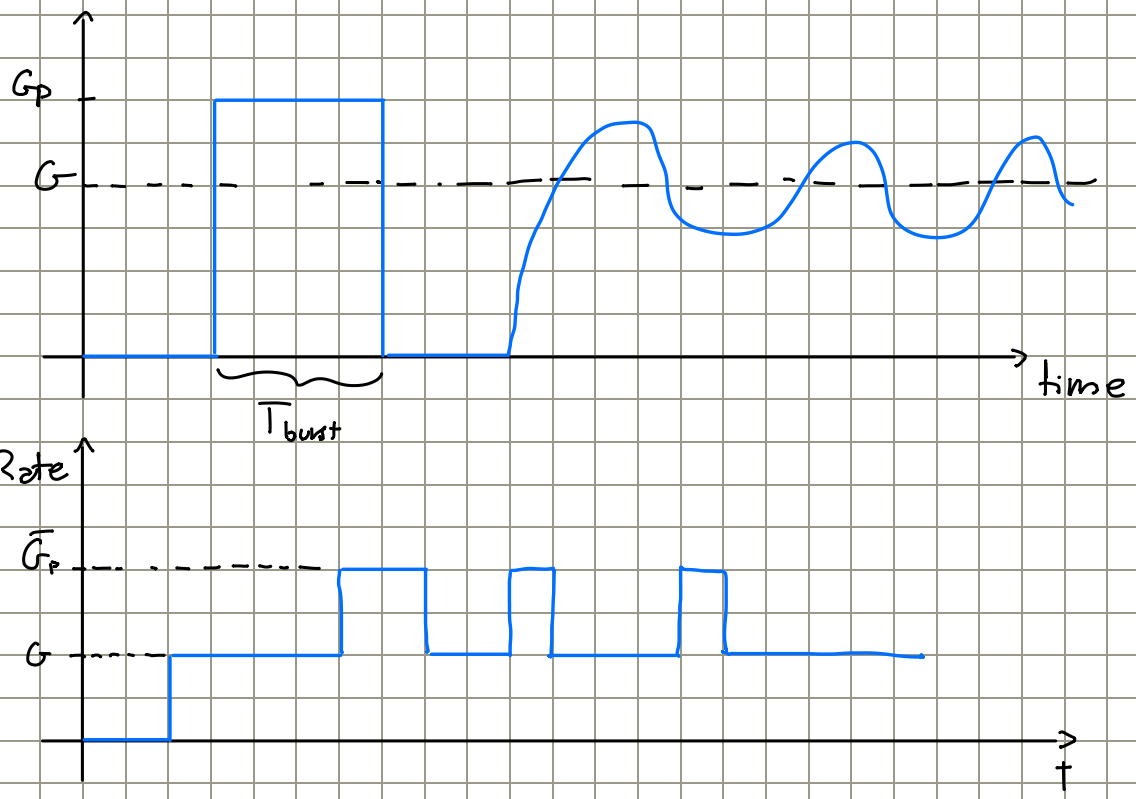
Initial bucket size: b(0) = b(full)

$$G = \frac{W \cdot B}{A} = \frac{80}{20}\ \text{Byte/s}$$

Suppose $\bar{G}$ = average traffic gen. rate [byte/s] = 100 Kbps

G_p = peak traffic gen. rate : $G_p > \bar{G} = 1$ Mb/s

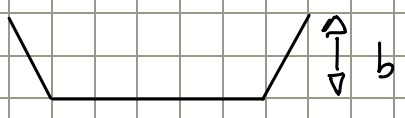
T_{burst} = max burst duration = 10^{-2} s



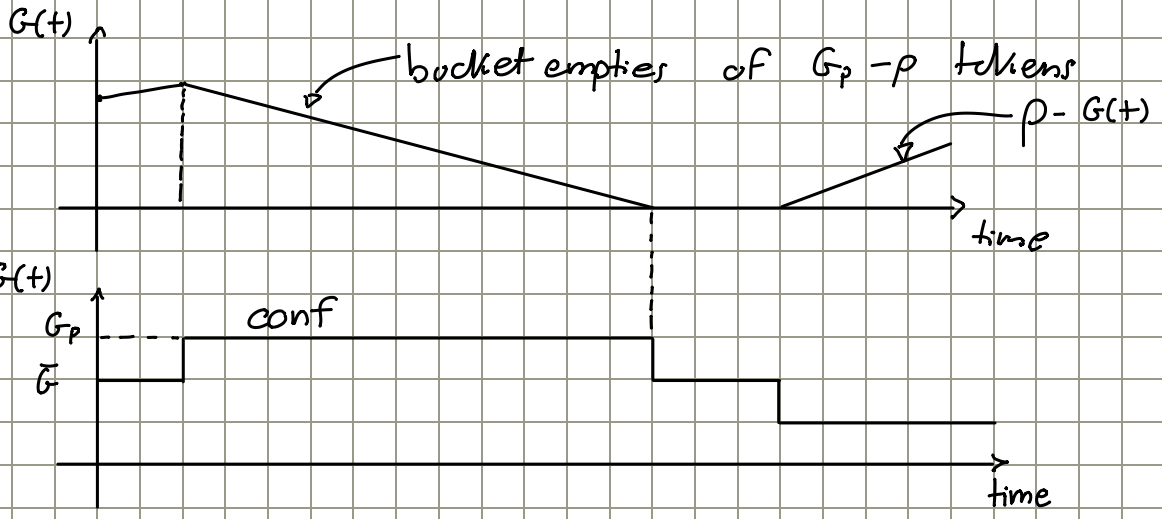
S.L.A. : $\rho = ?$

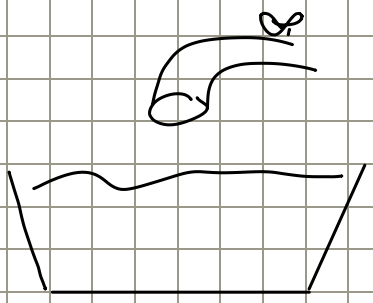
1 token → 1 byte $b = ?$

$\rho \geq$ average traffic gen. rate = $\bar{G} = 100$ $\frac{\text{K tokens}}{s}$



When source generates traffic at rate $G_p > \bar{G}$: it consumes tokens from the bucket at a rate larger than ρ





b: time to empty bucket when source transmits G_p is $\geq T_{burst}$



$G_p > \rho$

emptying rate: $G_p - \rho$

time to empty the bucket

$$G_p = \frac{b}{G_p - \rho} \geq T_{burst}$$

$\rho = \bar{G}$

$b = \overline{T}_{burst} \cdot (G_p - \rho)$

Start from an empty bucket

$b(0) = 0$

$\bar{G} = 100$ Kb/s

$G_p = 1$ MB/s $\Rightarrow$ $b = 800 \cdot 10^3 \cdot 10^{-2} = 8$ K tokens

$T_{burst} = 10$ ms

1) What is the maximum traffic rate the sure can generate at time $t=0$ remaining conformant?

2) If it generates a CBR traffic of rate = 1/2 of that point 1 after how many seconds can it tx at peak rate for T_{burst}?

3) If the rate of point (2) is maintained for $t=500$ s what will be the max burst period allowed to the source?